

Do Used Car Buyers Benefit from Electronic Vehicle Credits? A Study of U.S. Federal Credit

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Motivation/Research Questions

- Globally Increased CO2 emission
- Environmental problem and climate change issue
- Technology Development (HEVs)
- Policy Change (government changes the policy frequently)

Motivation/Research Questions (Cont'd)

- Expanded tax credit policy on HEV (Hybrid or Electric Vehicles)
 - Who benefits from credits for the purchase of used HEVs?

Why looking at the used vehicle market?

- Used car market in the US is big enough to study
- It accounts about 25% of total vehicle sales

Literature Review

- Sallee, James M. 2011. "The Surprising Incidence of Tax Credits for the Toyota Prius." *American Economic Journal: Economic Policy*, 3 (2): 189-219.
- Boyle, Melissa & Matheson, Victor. (2008). *Measuring Tax Incidence: A Natural Experiment in the Hybrid Vehicle Market*. *Environmental Economics and Policy Studies*. 10. 10.1007/BF03353981.
- Sumeet Gulati, Carol McAusland, James M. Sallee, Tax incidence with endogenous quality and costly bargaining: Theory and evidence from hybrid vehicle subsidies, *Journal of Public Economics*

Presentation Overview

1. Policy Structure
2. Data Structure
3. Estimation Methods/Strategy
4. Conclusion

Brief history of the HEV tax credit policy

- First introduced as part of the *Energy Policy Act of 2005*
- In 2009, the American Recovery and Reinvestment Act expanded the credit to plug-in EVs and increased the maximum credit.
- In 2023, expanded the eligibility to used HEV purchase

Brief history of the HEV tax credit policy

Table (policy history)

Date Effective	Tax Credit for New vehicles	Tax Credit for Used vehicles
2005 - 2009	max \$3,400	\$0
2009 - 2023	max \$7,500	\$0
2023 - present	max \$7,500	max \$4,000

Policy Structure

We are interested in 2023 tax credit policy...

Table

Category	Eligibility
Model	Certain type of HEV (See the note)
Date Effective	After Jan 1, 2023
Credit Amount	$\max(4000, 0.3P)$, where $p < 25,000$

Note: following link has all different types of eligible vehicles.
<https://fueleconomy.gov/feg/taxused.shtml>

What vehicle models are included?

Table

Models	Years	Purchase Date
Ford Fusion	2013-2020	Jan 1/2022 - Mar 31/2024
Ford Fusion Energi	2013-2020	Jan 1/2022 - Mar 31/2024
Volkswagen Golf	2015-2019	Jan 1/2022 - Mar 31/2024
Volkswagen e-Golf	2015-2019	Jan 1/2022 - Mar 31/2024
KIA Optima	2017-2020	Jan 1/2022 - Mar 31/2024
KIA Optima Hybrid	2017-2020	Jan 1/2022 - Mar 31/2024
Hyundai Sonata	2016-2020	Jan 1/2022 - Mar 31/2024
Hyundai Sonata Hybrid	2016-2020	Jan 1/2022 - Mar 31/2024

Data Structure

TABLE: Summary statistics

Variable	Obs	Mean	Std. dev.	Min	Max
CPIadjuste~e	4,914	19002.86	4905.032	8713.87	33628.3
credit	4,914	1088.086	1758.183	0	4000
post_2023_~V	4,914	.2777778	.4479488	0	1
HEV	4,914	.5	.5000509	0	1
age	4,914	284.7619	103.1223	104	572
post_2023	4,914	.5555556	.4969546	0	1
gas_price	4,914	3.678291	.4470478	3.053	5.006
i	4,914	21.5	12.12215	1	42
t	4,914	59	33.77719	1	117

(Need to be updated little bit)

Estimation Methods and Regression

Our goal is to...

- I. Find the effect of credit on vehicle prices
- II. Find how \$1 increase in credit is shared between buyers and sellers

What approach do we use?

For first, we use OLS and DiD method

For second, we use OLS regression and 2SLS method

Estimation Method

Regression Model

$$\text{price}_{it} = \beta_0 \text{HEV}_i + \beta_1 \text{post2023}_t * \text{HEV}_i + \beta_2 \text{age}_{it} + \lambda_t + \lambda_i + \varepsilon_{it}$$

Where i = different vehicles, t = time

HEV : dummy variable. 1 = Hybrid or Electric, 0 = Gasoline

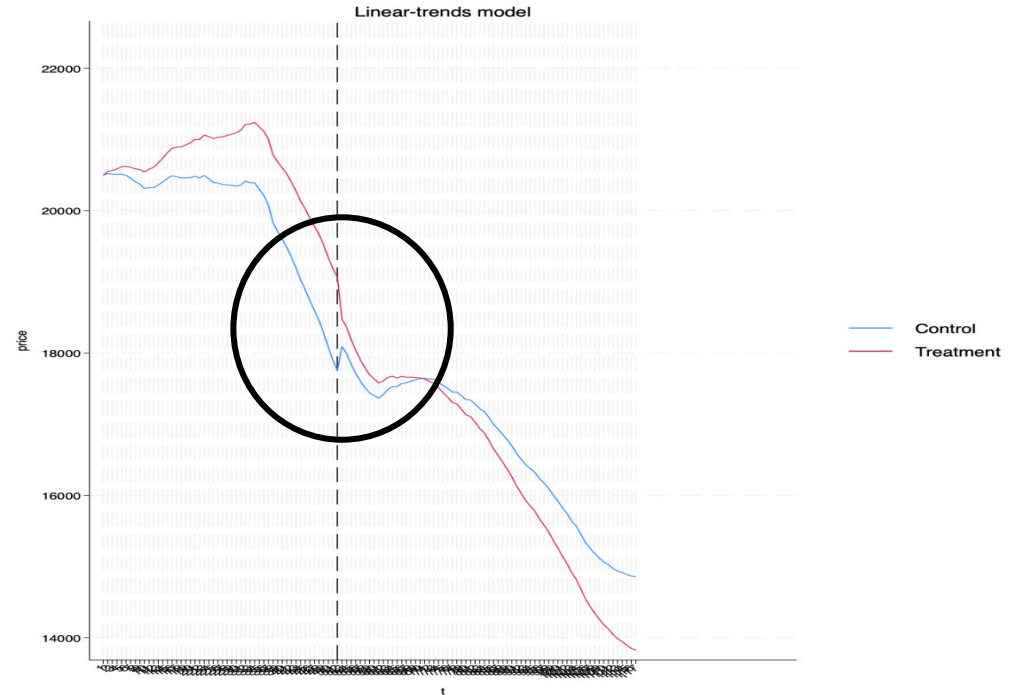
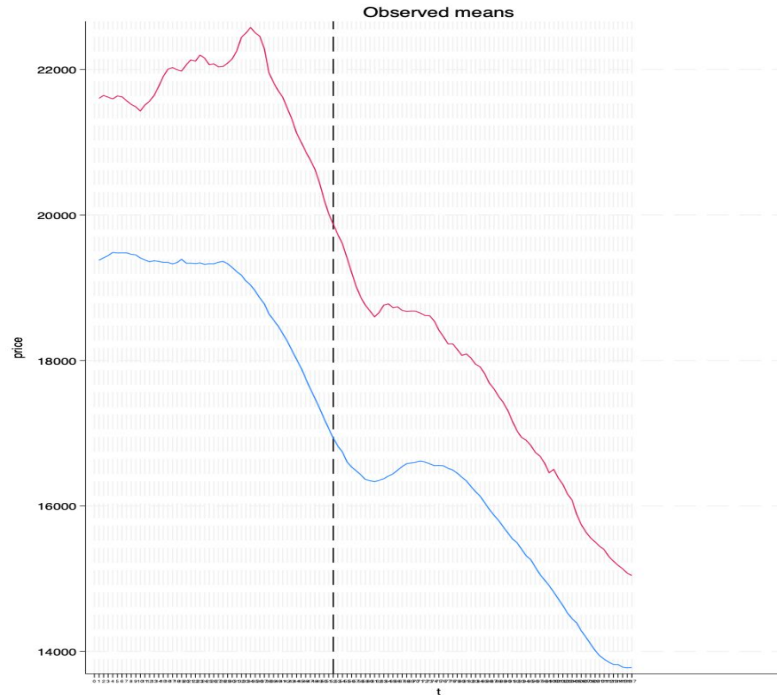
Age : age of cars

Ramda : i = vehicles fixed effect, t = time fixed effect

Difference and Difference Method

Graphical illustration of DiD model

Graphical diagnostics for parallel trends



DiD regression result

Negative relationship between vehicle price and policy change.

Table

	Diff-in-diff	Regression	Regression	Regression
		[1]	[2]	[3]
post_2023_HEV (1 vs 0)	-1120.547 (346.5184) ***	-5349.894 (295.119) ***	-1120.547 (346.5546) ***	-1120.547 (346.5546) ***
HEV	----	----	2967.202 (1382.75) ***	2967.202 (1382.75) ***
AGE	----	----	----	-37.92682 (1.846863) ***

OLS Regression Model

$$\text{price}_{it} = \beta_1 \text{credit}_{it} + \beta_2 \text{age}_{it} + \lambda_t + \lambda_i + \varepsilon_{it}$$

2SLS method

$$\text{price}_{it} = \beta_1 IV + \beta_2 \text{age}_{it} + \lambda_t + \lambda_i + \varepsilon_{it}$$

Three IVs

1. Gas Price
2. Total Energy emission
3. Coal CO2 Emissions

Economic intuition of these variables?



First Stage


$$\text{credit}_{it} = \beta_1 IV + \lambda_t + \lambda_i + \varepsilon_{it}$$

$$\text{credit}_{it} = \beta_1 IV + \lambda_t + \lambda_i + \varepsilon_{it}$$

Table

	Regression	Regression	Regression
credit	[1]	[2]	[3]
<u>gas_price</u>	-1179.759 (184.8181) ***	----	----
<u>TotalEnergyemission</u>	----	-5.178338 (.8102294) ***	----
CoalCO2Emissions	----	----	-36.51607 (5.721102) ***
cons	5427.582 (679.8148) ***	3198.104 (330.1442) ***	3647.332 (400.9661) ***

They all have statistically significant relationships with credit

Regression with the IVs

$$\text{price}_{it} = \langle \text{gas_price}_t \rangle + \text{age}_{it} + \text{fe} + \varepsilon_{it}$$

$$\text{price}_{it} = \langle \textit{CoalCO2Emissions}_t \rangle + \text{age}_{it} + \text{fe} + \varepsilon_{it}$$

$$\text{price}_{it} = \langle \textit{TotalEnergyemission}_t \rangle + \text{age}_{it} + \text{fe} + \varepsilon_{it}$$

Results from 2SLS

Table						
IV	<u>gas price</u>		CoalCO2Emissions		<u>TotalEnergyemission</u>	
price	[1]	[2]	[1]	[2]	[1]	[2]
credit	-3.016609 (.0839832) ***	-5.071514 (1.976248) ***	-2.966063 (.0851942) ***	-4.391816 (2.032383) ***	-1.829026 (.2134178) ***	-1.241167 (.2327253) ***
age	----	54.25723 (49.18122)	----	37.35829 (50.56399)	----	-40.97448 (5.820698) ***

After checking Underidentification test, Weak identification test and Sargan statistic (overidentification test of all instruments), the last column seems to be the strongest IV.

Results from 2SLS

Comparing with regular OLS...

(Std. err. adjusted for 42 clusters in i)

	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
credit	-1.3605	.0682934	-19.92	0.000	-1.498421	-1.222578
_cons	20483.2	74.30913	275.65	0.000	20333.13	20633.27
sigma_u	4775.6299					
sigma_e	2092.6164					
rho	.83892098	(fraction of variance due to <u>u_i</u>)				

- \$1 increase in credit decreases vehicle price by about \$1.5

Conclusion

- The first regression result shows that price of used HEVs decreases after the policy was introduced
- The second regression result shows that buyers capture the benefit of tax credit
- This credit policy may be appropriate policy to encourage consumers to purchase HEVs

Problems/Challenges

- There may be better/stronger IV that I can use
- I may need to include more variables
- Creating clear tables and graphs
- Interpretation of the result
- Mistakes on regression methods

Q & A

Thank you!